

Benchmark Report: pyflate

A small link budget, a compressed payload, and a lot of work between the bits

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Somewhere between a spacecraft and a ground station there is a link budget measured in kilobits.

Compression makes the most of it; decompression turns the arriving bits back into useful data. This benchmark gives that quiet half of the job to Python: read a bit stream, recover symbols, and undo several transformations. Its Huffman matcher looks like the obvious place to start. Measuring it led us to improvements across the whole decoding pipeline, then a Rust extension that executes the symbol decoder through one call per block.

1. Workload and result

Despite its name, the measured pyflate workload is **bzip2**, not DEFLATE. Each iteration decompresses `interpreter.tar.bz2` from **67,562 bytes to 399,360 bytes**; the harness checks the original MD5 outside the timer. The Python tier uses the standard library rather than calling a native decompression library. Its main structures are an integer bit buffer, Huffman tables, a move-to-front (MTF) alphabet, Burrows-Wheeler transform (BWT) index vectors and output buffers.

One block, six Huffman tables, 148,271 symbols. The decoder switches tables every 50 symbols. Huffman decoding, move-to-front and RUNA/RUNB expansion produce a 336,184-byte L-vector. Inverse BWT and final RLE4 expansion produce the 399,360-byte output.

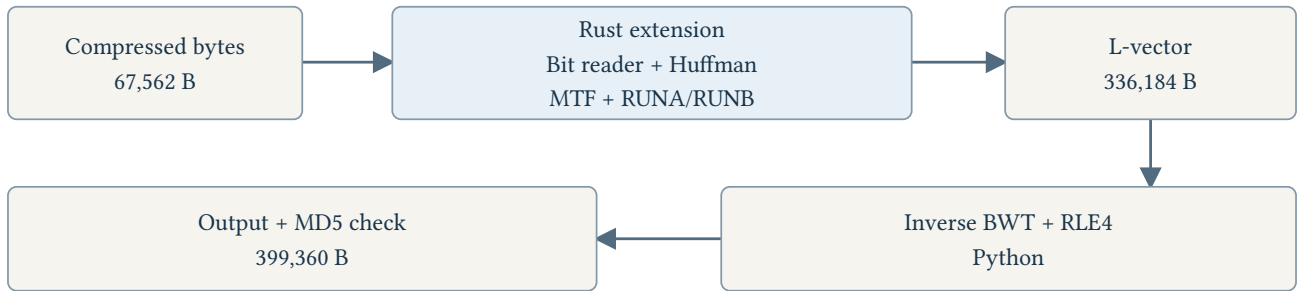


Figure 1: The bzip2 path. RUNA/RUNB and move-to-front are handled together in the symbol loop. The native kernel covers the same combined boundary proposed for hardware.

Configuration	Mean ± SD	vs stock
Stock Python	1,129.89 ± 12.68 ms	1.00×
Optimized Python	288.00 ± 3.22 ms	3.92×
Python + Rust extension	173.59 ± 2.15 ms	6.51×

Our final software path is 6.51× faster than stock: an 84.6% runtime reduction. Python improvements provide 3.92×; the Rust symbol decoder adds 1.66× over that optimized baseline. Both tiers exceed the course’s 7% improvement requirement and match `bz2.decompress` byte for byte, passing the unchanged MD5 check. Rust handles the blue stage above; block headers, inverse BWT and RLE4 remain in Python.

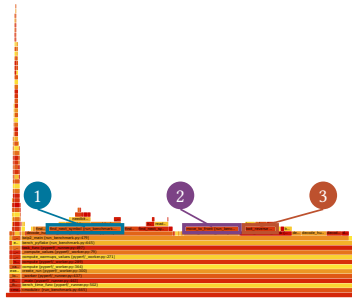
Measurement scope. Course QEMU/KVM VM, Ubuntu 22.04, release CPython 3.10.12, pyperf 2.10.0; 120 measured values per configuration, collected sequentially on guest CPU 0. The current Rust source was rebuilt and checked on this VM. Source: `results/vm_release_20260907/pyflate_{stock,python,native}.json`. Development profiles and ablations are labeled separately.

Medians are 1,128.01 ms, 287.84 ms and 172.91 ms, with interquartile ranges of 12.97 ms, 3.62 ms and 2.35 ms – about 1.1%, 1.3% and 1.4% of the median in each case. Unlike `nbody`, these three runs are not dominated by between-worker variation: the optimized Python tier shows no detectable worker effect at all (ICC 0.00) and the other two are moderate (0.58 and 0.22), so their 120 values are worth 56 to 120 independent observations rather than about 40. Appendix A5 gives the decomposition and the corrected standard errors.

2. Profiling: the whole decoding pipeline

The stock Python-frame profile highlights Huffman decoding and bit-buffer work, with move-to-front and inverse BWT alongside them. Development instrumentation found a mean of about five entries visited by the symbol matcher, despite its table capacity of 258. A lookup table can remove this scan, but the measured scan is short; per-symbol interpreter work and the later transformations also deserve attention.

Full recorded flame graph

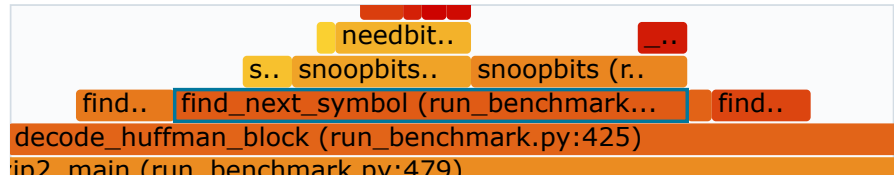


Original widths and stack depth.
Imports and harness retained.
Each outline marks one call path.
The same function may recur.
Percentages are inclusive,
not flat self-time totals.

Highlighted regions - enlarged with surrounding frames

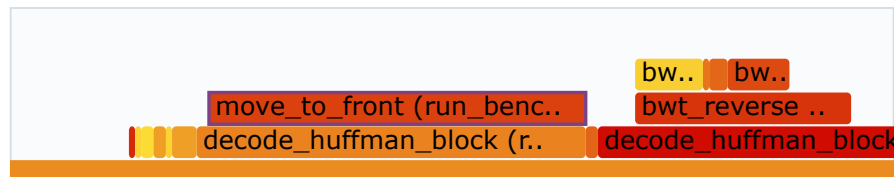
1. Huffman matcher

find_next_symbol | this frame: 20.54% inclusive



2. Alphabet updates

move_to_front | this frame: 15.10% inclusive



3. Inverse BWT

bwt_reverse | this frame: 8.66% inclusive

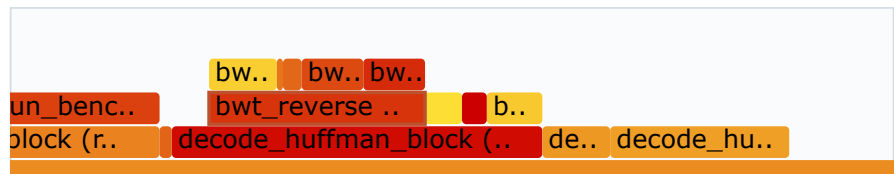


Figure 2: Full stock Python-frame flame graph, retaining startup and harness context. Numbered outlines identify exactly the call paths enlarged on the right. Inclusive percentages use the original whole-profile denominator; nested shares overlap.

Function	Stock self	Stock incl.	Opt. self	Opt. incl.
decode_huffman_block	23.76%	93.81%	37.50%	93.27%
find_next_symbol	16.34%	40.59%	–	–
move_to_front	15.10%	15.10%	–	–
snoopbits	9.65%	17.82%	–	–
readbits	6.44%	7.43%	–	–
_mask	2.97%	2.97%	–	–
bwt_reverse	4.70%	13.37%	25.96%	50.00%
bwt_transform	8.66%	8.66%	24.04%	24.04%
rle4_expand	–	–	4.81%	4.81%

Self is time in the function itself; **inclusive** is time in it and everything it calls. Shares are of each profile's own total and are not comparable between the two columns, which are independently normalized – the optimized run is 3.92× shorter in absolute time. A dash means the function no longer exists after the rewrite: the bit reader and the symbol matcher were folded into `decode_huffman_block`, and `move_to_front` became an inline `l.append(l.pop(-r))`.

The table aggregates every call site of a function; the figure's highlight percentages describe **one** outlined frame, which is why `find_next_symbol` reads 20.54% there and 40.59% here. Regenerate both from the same recorded SVGs the figures use with `report/summarize_profiles.py`.

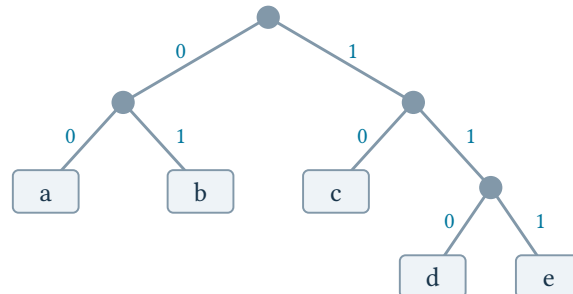
The table is the argument for what shipped. In the stock profile the symbol loop and its bit reader account for the bulk of the work, and the inverse BWT is a distant third. After the rewrite the ordering inverts: `decode_huffman_block` still dominates self time, but what remains beside it is `bwt_reverse` and

bwt_transform – which is why the Rust kernel is drawn around the symbol decoder and why the remaining Amdahl ceiling is the BWT.

A separate **Windows / CPython 3.12.6** cProfile run gives find_next_symbol 0.519 s cumulative out of 1.186 s (43.8%), including its bit-reader callees. decode_huffman_block has 0.241 s **self** time and 1.172 s **cumulative** time: most of its inclusive cost is in functions it calls. Source: dev/pyflate/FINDINGS.md §3. That development run and the VM profile above agree on the ordering, not on the numbers; different machine, different interpreter, different estimator.

Example codes

a = 00
b = 01
c = 10
d = 110
e = 111



Canonical ranges

len 2: 00..10

len 3: 110..111

Match a length range,
then index the symbol.

Example: 110 decodes to d.

Figure 3: Illustrative canonical Huffman codes. Codes of equal length are consecutive; length-indexed ranges identify the symbol without scanning individual candidates. The optimized implementation adds a primary lookup for short codes.

3. Changes that shipped

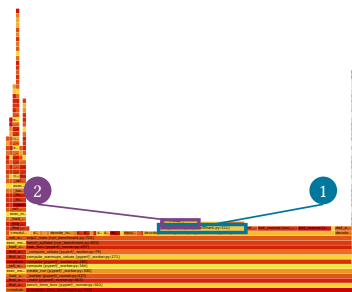
- **Reduce per-byte overhead:** read the compressed input once, refill the bit window in chunks, bind hot values to locals and accumulate bytes without allocating an object per byte.
- **Decode canonical Huffman codes:** use length-indexed tables and a primary lookup table with a fallback for longer codes. Table construction is amortized over the block.
- **Improve the back end:** replace comparison sorting with counting-sort BWT construction; reverse the MTF list so updates move $O(\text{rank})$ entries; use regex-assisted RLE4 expansion to reduce Python loop iterations. The byte scan/output work is still linear overall.

Component reverted from T3	VM trial 1	VM trial 2	Dev, 3.12
None (full T3 runtime)	271.2 ms	273.9 ms	175.5 ms
Regex-assisted RLE4	+100.6 ms	+98.7 ms	+32.3 ms
Primary Huffman lookup	+52.6 ms	+49.4 ms	+10.1 ms
Counting-sort BWT	+20.8 ms	+17.2 ms	+38.6 ms

Ablation scope. Each row puts one stock component back into the optimized Python decoder (dev/pyflate/ablate.py, which drives the T3 module directly rather than the pyperf harness) and reports the added time, best of seven interleaved decompressions. Costs need not add. The two VM trials are separate runs on the course VM, release CPython 3.10.12, pinned to one guest CPU; the last column is the earlier development run on Windows / CPython 3.12.6 (dev/pyflate/FINDINGS.md §2b).

The measured platform reverses part of the development ranking. On the VM, regex-assisted RLE4 is worth about 100 ms and the primary Huffman lookup about 50 ms – both more than counting-sort BWT at about 20 ms – and the two trials agree within 4 ms on every row. The development run had ranked BWT first and the Huffman table last, and an earlier version of this report concluded from it that every back-end change outweighs the matcher table. On the course VM that is false: the table is the second-largest single contribution. What survives on both platforms is that no single change dominates, so the pipeline, not the matcher alone, had to be optimized. Raw output: results/vm_rerun_20260910_3697a63/.

Full recorded flame graph

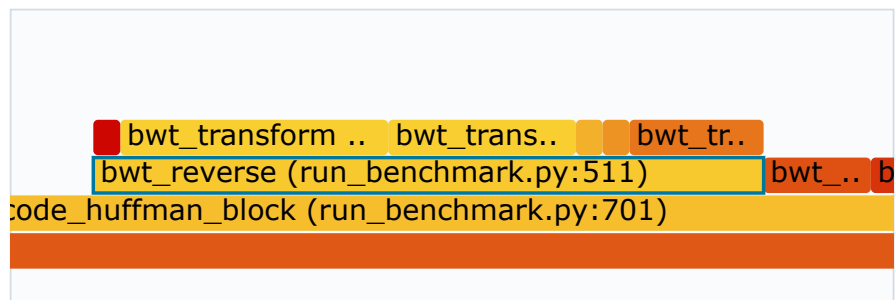


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Highlighted regions - enlarged with surrounding frames

1. Inverse BWT

bwt_reverse | this frame: 24.04% inclusive



2. Index-table construction

bwt_transform | this frame: 9.62% inclusive

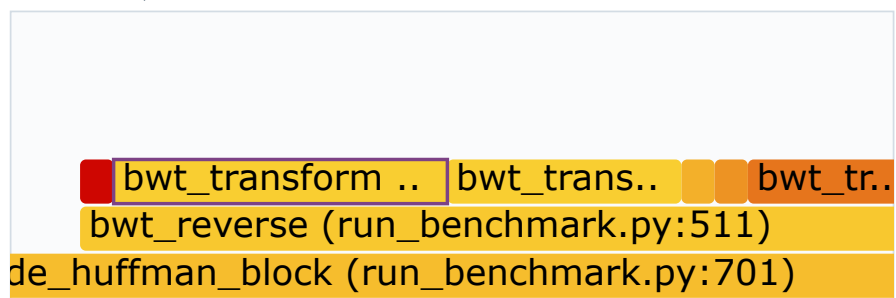


Figure 4: Full optimized **Python** profile, before native offload. Inverse BWT and its index-table construction remain visible. Matcher work was inlined, so its old function frame disappears without implying that Huffman decoding is free.

4. Native execution: what improves, what remains

Our Rust/PyO3 BlockDecoder handles the bit reader, Huffman decode, MTF and RUNA/RUNB expansion. Header parsing, inverse BWT, RLE4 and MD5 remain in Python. One block-level call returns the L-vector and the ending bit position; checking both catches output errors and stream desynchronization.

Same benchmark file	Mean $\pm$ SD	vs Python
Python backend	288.00 $\pm$ 3.22 ms	1.00×
Native decode backend	173.59 $\pm$ 2.15 ms	1.66×

These are the same 120-value measurements summarized on page 1, covering the complete benchmark. The new VM build includes the submitted Rust refactor. `HWSW_BACKEND=python|native|auto` selects the implementation; native mode requires the extension, while auto mode falls back. Pyperf workers need `--inherit-environ HWSW_BACKEND`.

The submitted Rust source was rebuilt on the VM. Ten Rust unit tests and seven blocks across five Python/bz2 fixtures passed, including exact ending bit positions. A dispatch check confirmed that native and auto load the new extension and call it once per block. The crate separates Python bindings, bit reading, Huffman tables, block decoding and trace serialization; the Python API is unchanged. After installing the current build, `script_pyflate.sh native` runs this tier.

Matched-work CPU counters

These are medians of three warm-loop VM runs, each decoding the input 16 times, divided by 16. Both backends produced the same output digest. They are a separate experiment from the rigorous timing table above, with acknowledged counter gating after setup (Appendix A2).

Metric per complete decode	Python	Hybrid native
Elapsed time	288.68 ms	174.44 ms
Instructions	1,947.94 M	1,171.65 M
Cycles	675.92 M	404.24 M
IPC	2.88	2.90
Branches	349.43 M	207.30 M
Branch misses (rate)	544,260 (0.156%)	301,458 (0.145%)
Generic cache references	1.749 M	1.686 M
Generic cache misses (rate)	69,322 (3.964%)	52,842 (3.133%)

Native decoding reduces instructions by **39.9%**, cycles by **40.2%** and branch misses by **44.6%**, while IPC changes little. The approximately 1.65× speedup in this experiment comes mainly from removing interpreter work. Generic cache misses fall **23.8%**, a smaller reduction than instructions: the remaining Python BWT/RLE4 pipeline still processes the whole block. Aggregate counters cannot attribute misses to a particular stage. L1 events are omitted because an earlier capture returned invalid load counts.

Offload limits and the next target

Amdahl's law gives $S = 1 / ((1 - f) + f/s)$ for kernel speedup s and offloaded fraction f of **optimized Python** runtime. Even infinite symbol-decoder speed leaves BWT and RLE4. Unlike nboddy's native integration loop, this offload leaves over a billion instructions per decode in a largely Python pipeline. The next target is the **whole inverse BWT**, including table construction, not just its dependent traversal. Earlier VM phase timings put them at 137.7 ms and 47.8 ms respectively (Appendix A3). A native port and working-set sweep would test whether memory latency then becomes limiting; the present data do not establish that.

5. Hardware acceleration: current proposal

The hardware boundary comprises `huffman_engine` followed by `mtf_cam`. Together they produce the same L-vector as the native decoder. Software keeps block headers, inverse BWT, RLE4 and MD5. A regular decoder datapath and a coarse block-level submission are the main co-design opportunities.

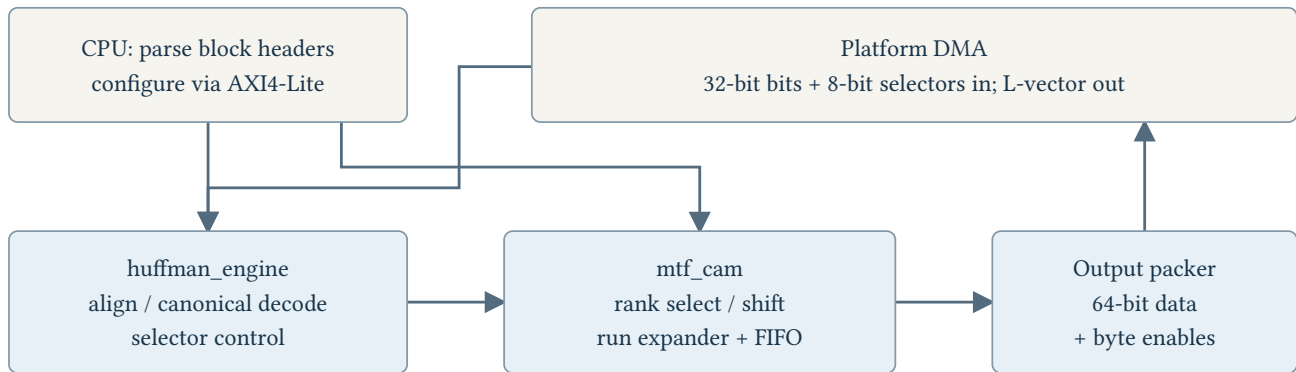


Figure 5: AXI4-Lite configures both modules; AXI4-Stream carries data. The symbol stream stays on chip. Platform DMA returns the L-vector to software.

Huffman engine

A bit aligner feeds a canonical-code comparator/lookup datapath. The selector controller changes the active table every 50 symbols. The current interface specifies a 32-bit compressed input stream, an 8-bit selector stream, and 32-bit output beats containing a 9-bit bzip2 symbol plus type fields. A 32-bit AXI4-Lite window holds controls, `START_BIT`, lengths and counters. It supports up to six tables and a maximum bzip2 code length of 20 bits.

The target is at most 1.10 cycles per symbol, including table construction and switches, with an unstalled benchmark-block model of 153,121 cycles. At the **50 MHz clock target**, that model is 3.06 ms for this module. This is not an achieved frequency or a measured end-to-end hardware runtime.

MTF and run expansion

The input is a **rank**, so the decoder selects an indexed alphabet entry and shifts preceding entries. The current proposal maintains up to 256 entries and expands RUNA/RUNB groups through a buffered output packer. The default output width is eight bytes (64-bit data plus byte enables). It targets one MTF symbol per cycle for any valid rank.

Measured mean rank 7.17 characterizes this input; the one-symbol-per-cycle requirement also covers large ranks. The specified area/throughput sweep varies output width $W = 4, 8, 16$ and FIFO depth. The default module model is 157,560 cycles, or 3.15 ms at 50 MHz. The two modules stream concurrently; their individual modeled latencies cannot simply be added or substituted for an integrated back-pressure simulation.

Implementation status. Requirements and architecture are approved for both modules.

Microarchitecture, RTL, verification, PPA and integration are still open in the recorded hardware status.

Formal MTF invariants are planned, not completed proofs. Drivers and platform DMA remain integration work. No measured area, clock or power benefit is claimed.

6. Conclusion

The delivered Python-plus-Rust path decompresses the input in **173.59 ms**, achieving **6.51× over stock** with byte-exact output. Python improvements provide 3.92× and the native decoder adds 1.66×, leaving BWT and RLE4 as the next software targets. The hardware proposal follows the same block-level interface; its integrated performance and cost remain to be demonstrated.

Evidence: `hw/huffman_engine/docs/{prd,m}as.md`, `hw/mtf_cam/docs/{prd,m}as.md`, and `hw/PROGRESS.md` (2026-09-05). Methods and supporting profiles: **report_appendix.pdf**. Repository: [HWSW_Project](#).